esprit
Ecole Supérieure Privée
d'Ingénierie et de Technologies

Restitution 1

LittleBig Connection

Implementation of an End-to-End Test Automation Chain with Centralized Reporting for a B2B Platform

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QA / Test Automation · May 2026

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Esprit Night class

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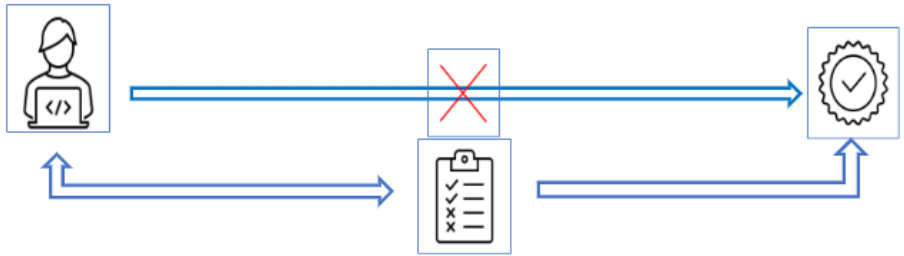
6 Realisation (Later Restitution 2

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Context & Host Organization

Contexte

- Frequent updates to mobile, web and desktop applications
- Any change can reduce software quality
- Increase of unexpected bugs
- Testing is required to ensure quality



Contexte



- ✓ Reduce time execution
- ✓ Ensure Software quality

Context & Host Organization

Mantu Group

Host company — Amaris Tunisia

2002	12K+	60+
Founded Geneva, Switzerland	Consultants worldwide	Countries global presence

- International independent consulting group
- Specializes in Technology, Management & Transformation
- Deploys consultants embedded inside client teams
- Operates in Tunisia under the Amaris brand
- Located at Technopole El Ghazala, Aryanah



Context & Host Organization

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B2B SaaS Platform · Mantu Group subsidiary · Founded 2012, Paris

Client

- Posts service requests
- Creates Purchase Orders
- Reviews proposals

Supplier

- Responds to RFPs
- Submits invoices
- Manages missions

Admin

- Validates PO flows
- Configures platform
- Manages users

What LBC does

RFP-based PO	Client posts request → Supplier responds → Admin validates → PO created
Quotation PO	Client requests a quote directly from a known supplier
Standalone PO	Client creates a Purchase Order directly without RFP

Team Organization

Squad Structure

per product squad

PM	Product Manager	× 1
TL	Tech Lead	× 1
UI	UI Designer	× 1
QA	QA Engineer	× 1
DEV	Developers	× 3 to 5

My role — QA Engineer embedded in the Contract Squad

QA Department

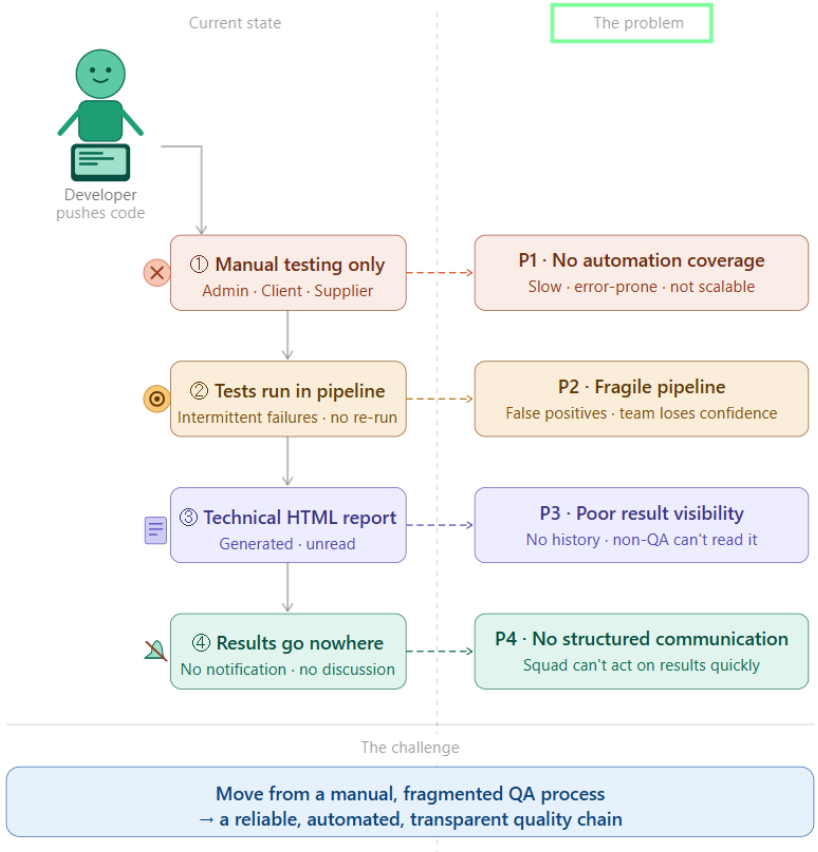
cross-squad team

MGR	QA Manager	× 1
TQL	QA Tech Lead	× 1
QA	QA Engineers	× 6

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Problem Statement

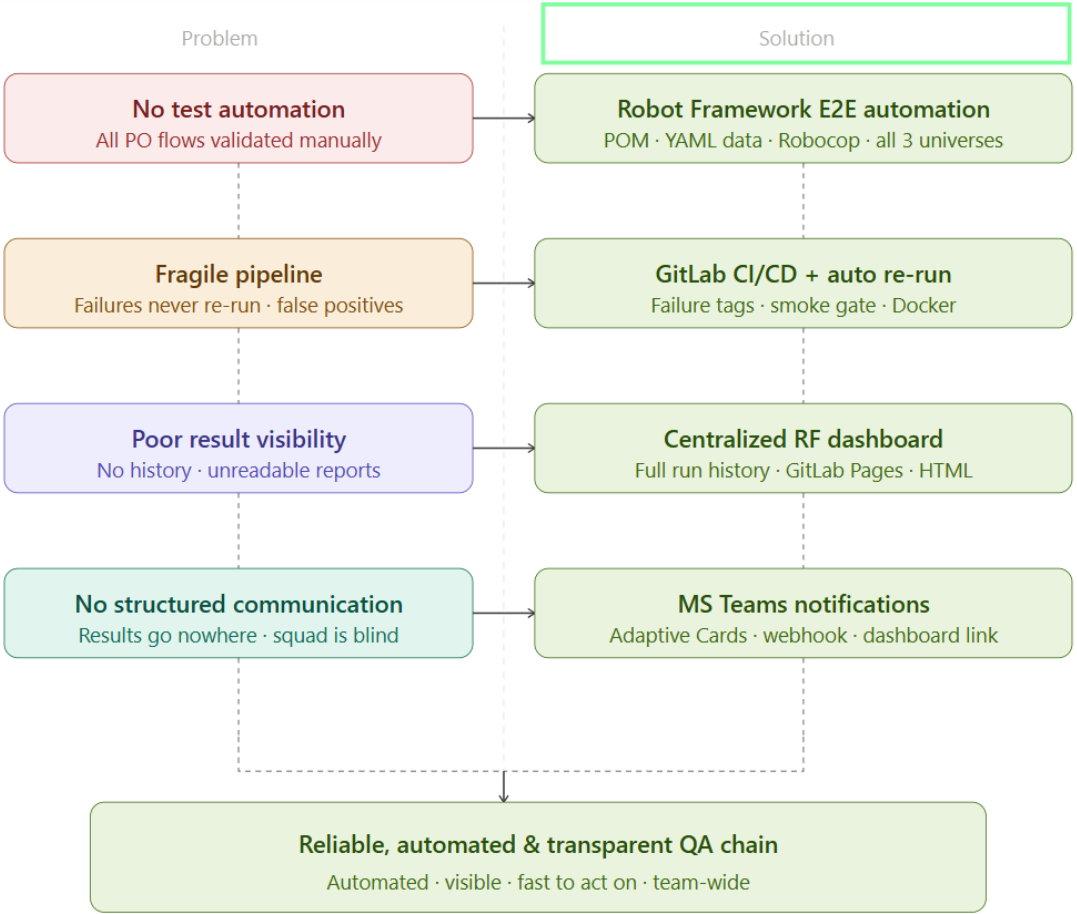
Problem Statement



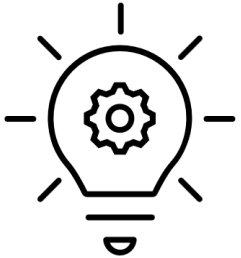
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Proposed Solution

Proposed Solution



Proposed Solution



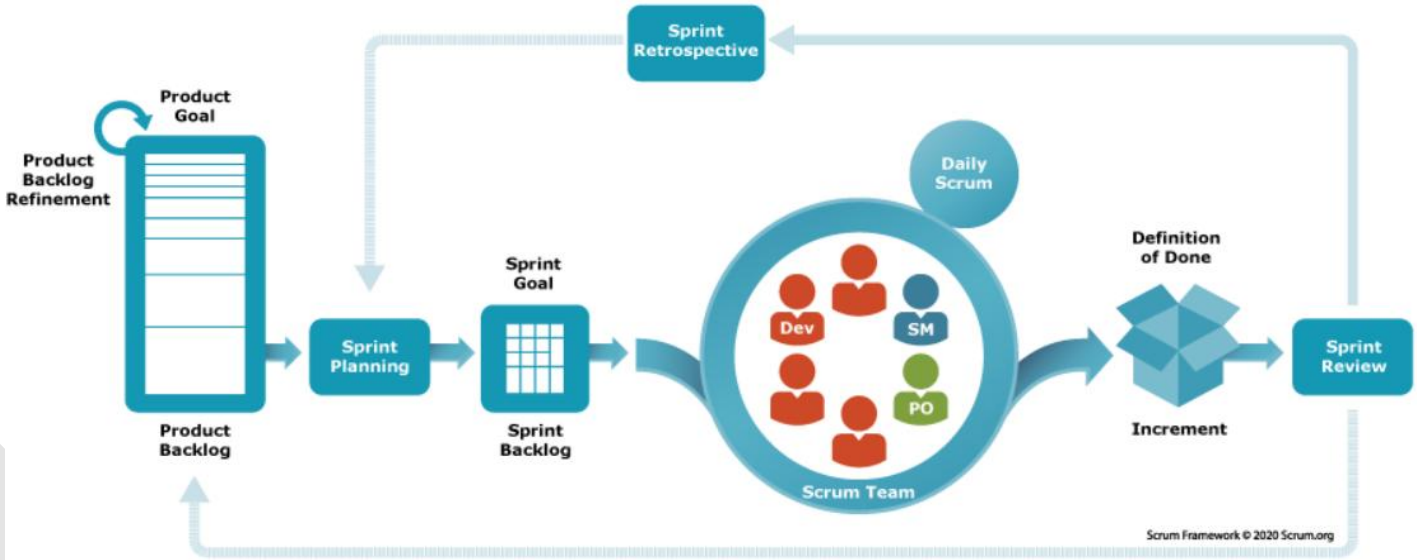
1. **Robot Framework E2E** : Automates all critical PO flows across Admin, Client and Supplier using POM, YAML data and Robocop.

2. **GitLab CI/CD + auto re-run** Tests run in Docker on every push. Failed tests are automatically re-run via failure tags to eliminate false positives.

3. **Centralized RF dashboard** A dashboard is generated after each run with full history, auto-deployed on GitLab Pages for team-wide access.

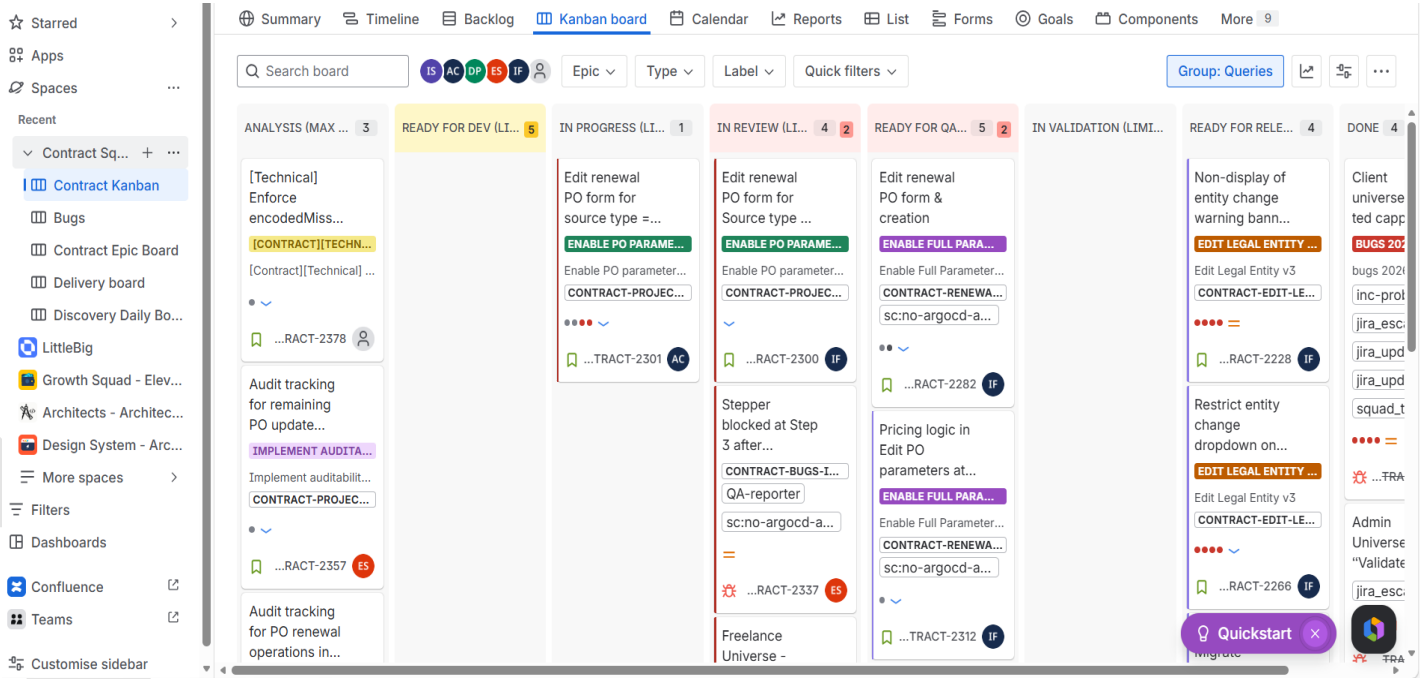
4. **MS Teams notifications** An Adaptive Card is sent to Teams after every pipeline with a run summary and a direct link to the dashboard.

Adopted Methodologies (Agile – Scrum)



Proposed Solution

Adopted Methodologies (Agile – Scrum)



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System Functional and Non-Functional Requirements

Functional Requirements

ID	Requirement	Priority
FR-01	Explore the LBC platform and analyze the existing test project structure	P1
FR-02	Write and execute Xray test cases for every Contract Squad new feature	P1
FR-03	Automate Admin, Client and Supplier PO flows (contract Purchase Orders) using Robot Framework	P1
FR-04	Integrate Contract Squad tests into the GitLab CI/CD pipeline	P1
FR-05	Automatically re-run failed tests via failure tags inside Docker	P1
FR-06	Generate a report dashboard with full execution history after each pipeline run, auto-deployed on GitLab Pages for team-wide access	P1
FR-07	Send a run summary with a dashboard link to MS Teams after each pipeline run	P2

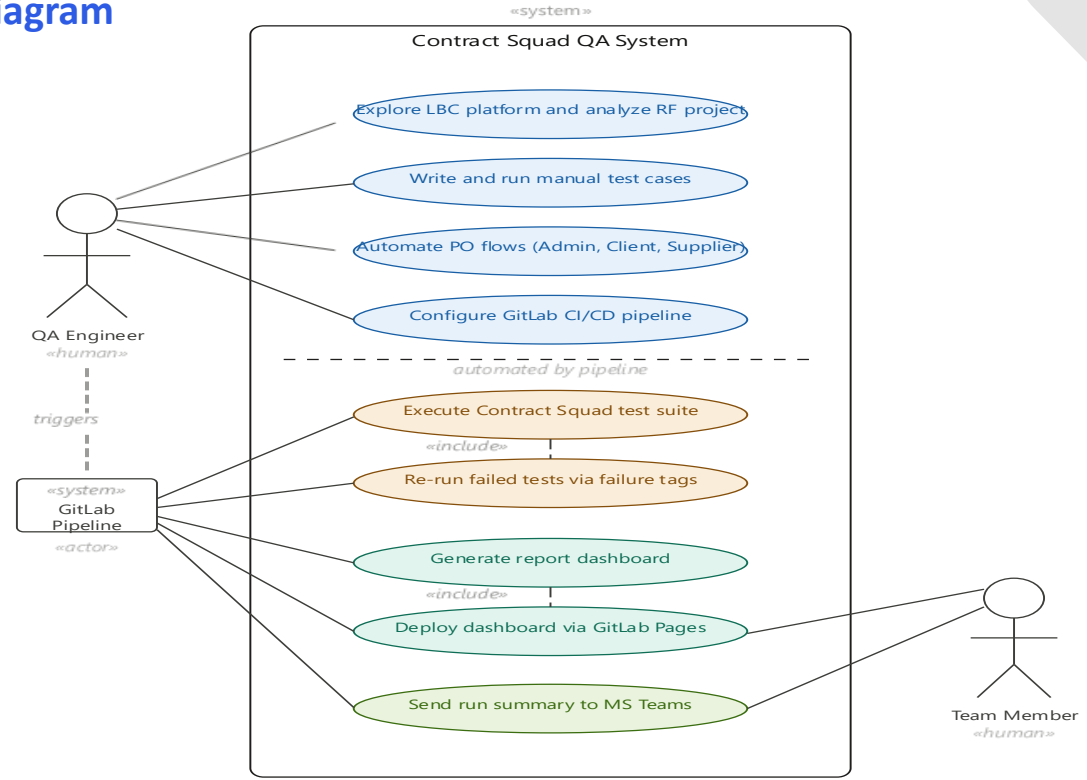
Non-Functional Requirements

ID	Requirement	Target
NFR-01	**Performance** Test suite completes within a reasonable time on the GitLab pipeline	GitLab pipeline time budget
NFR-02	**Reliability** Low false failure rate via automatic re-run	Failure tags strategy
NFR-03	**Maintainability** Code follows Page Object Model and passes Robocop linter	Code review validation
NFR-04	**Traceability** Results linked to Jira/Xray for the Contract Squad scope	Xray integration

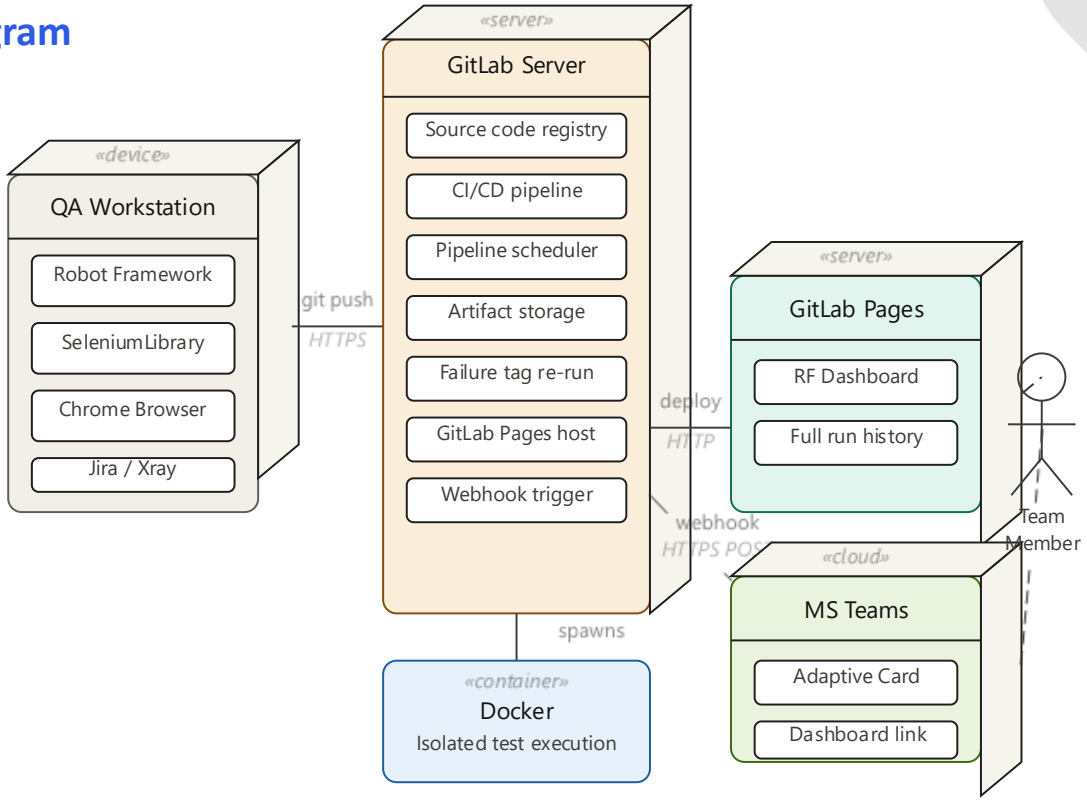
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Conceptual Design and Technologies

Global Use Case Diagram

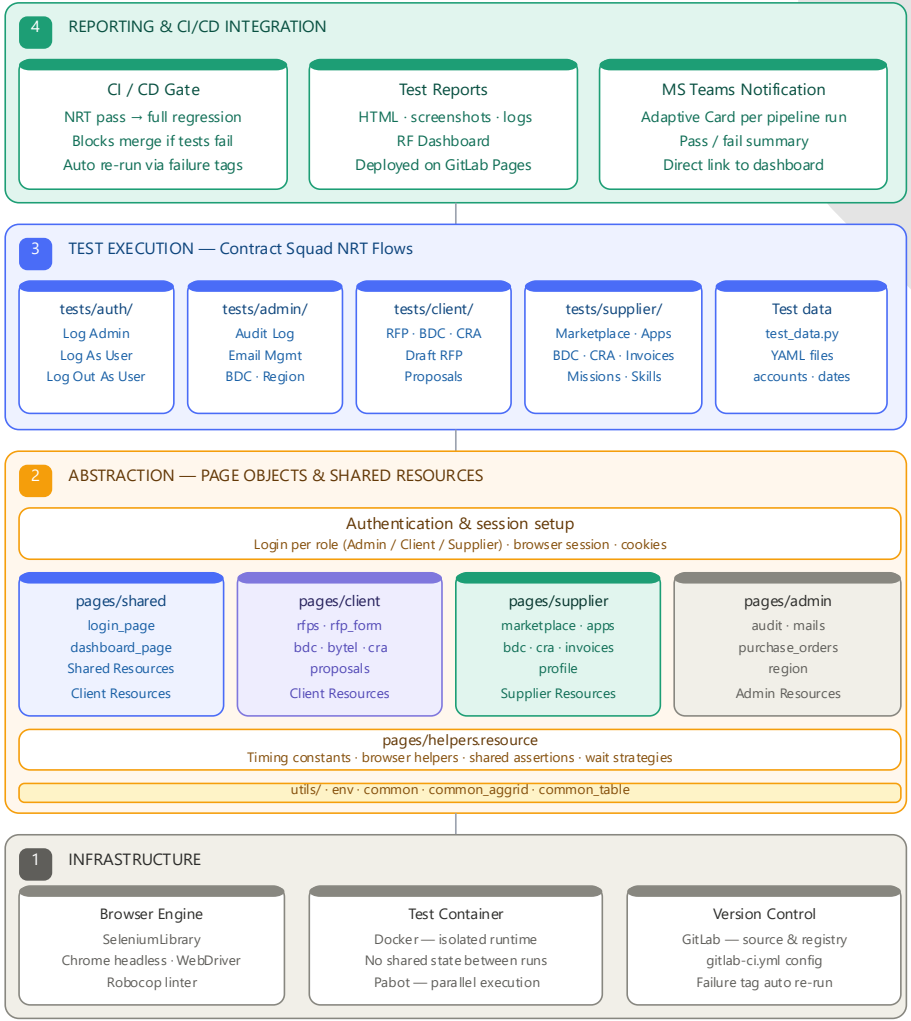


Deployment diagram

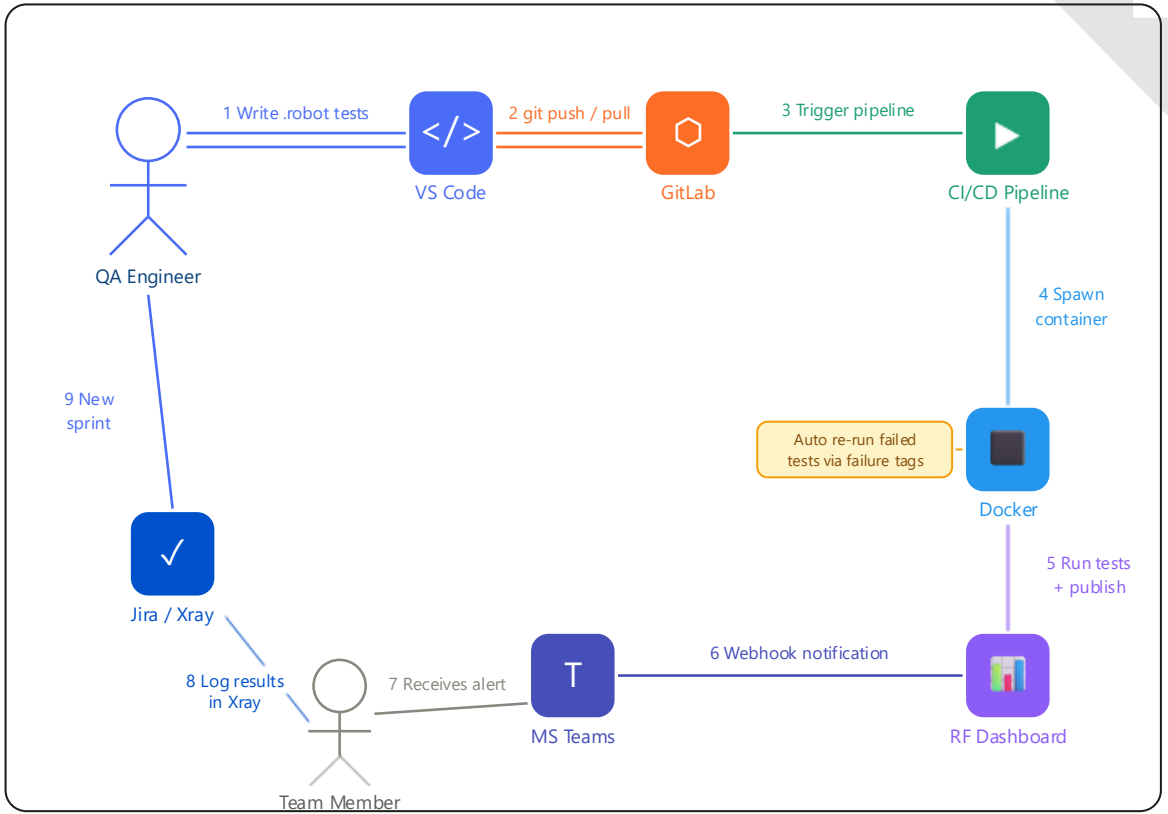


Conceptual Design and Technologies

Framework architecture layers



Workflow - Cycle



Technologies

- Python
- CI/CD : GitLab CI/CD
- Docker
- Selenium Library
- SQLite
- Robot Framework

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Realisation

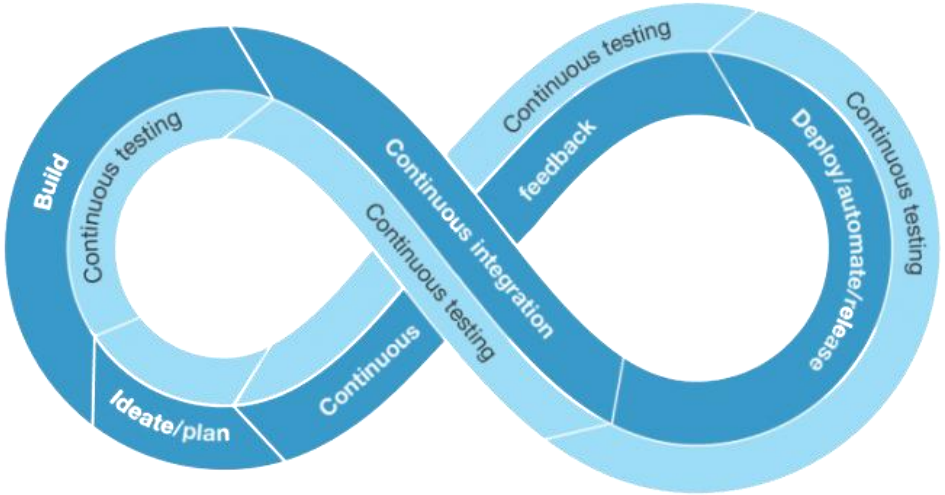
Thanks !

Do you have any question?

Imen Sebteoui
QA / Test Automation — LBC

➢ Méthodologies adoptées

- L’approche Dev Test Ops:



Functional Requirements

ID	Requirement	Priority
FR-01	Explore the LBC platform and analyze the existing test project structure	P1
FR-02	Write and execute manual test cases for the Contract Squad scope	P1
FR-03	Automate Admin, Client and Supplier PO flows (contract Purchase Orders) using Robot Framework	P1
FR-05	Integrate Contract Squad tests into the GitLab CI/CD pipeline	P1
FR-06	Automatically re-run failed tests via failure tags inside Docker	P1
FR-07	Generate a report dashboard with full execution history after each pipeline run	P2
FR-08	Auto-deploy the dashboard via GitLab Pages for team-wide access	P1
FR-09	Send a run summary with a dashboard link to MS Teams after each pipeline run	P2